

Task 5: Exploratory Data Analysis (EDA) on Titanic Dataset

Name:

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Objective

To perform Exploratory Data Analysis (EDA) on the Titanic dataset and extract meaningful insights using statistical summaries and visualizations.

Dataset Overview

The Titanic dataset contains information about passengers such as age, gender, passenger class, fare, and survival status.

Dataset Information

- Number of records: 891
- Features include:
 - Age
 - Sex
 - Passenger Class (Pclass)
 - Fare
 - Embarked
 - Survival Status

Statistical Summary:

- `describe()` was used to obtain statistical measures.
- `info()` was used to understand data types and missing values.
- Missing values were found in Age and Deck columns.

Visualizations and Observations

1. Histogram of Age

Observation:

Most passengers were between 20 and 40 years old.

2. Boxplot of Fare

Observation:

Several outliers are present in the Fare column, indicating some passengers paid significantly higher fares.

3. Correlation Heatmap

Observation:

Fare and Passenger Class show noticeable relationships with survival.

4. Pairplot

Observation:

Passengers in higher classes and those paying higher fares had better survival chances.

Key Findings

1. Female passengers had a higher survival rate than males.
2. First-class passengers survived more often than lower-class passengers.
3. Fare shows a positive relationship with survival.
4. Age has a weak effect on survival.
5. Several outliers exist in the Fare column.
6. Missing values were observed in Age and Deck columns.

Conclusion

The analysis shows that gender, passenger class, and fare significantly influenced survival chances on the Titanic. EDA helped identify patterns, trends, and anomalies within the dataset.
